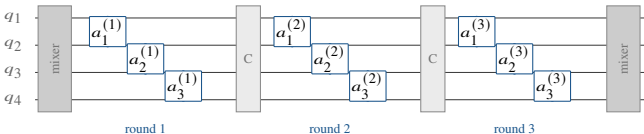


## (a) One QAOA cost layer, as a downstream compiler sees it

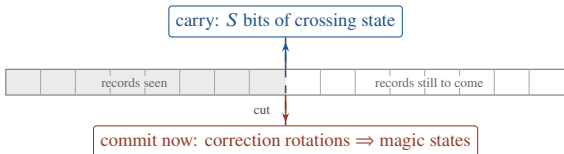
semantic  $e^{-i\gamma C}$ ,  $C = \sum_{j \leq m} w_j Z_{u_j} Z_{v_j}$   $m$  records ( $Z_{u_j} Z_{v_j}$ ,  $\gamma w_j$ )  
 one  $R_{ZZ}$  per edge  
 ↓ schedule + twirl + lower  
 one contiguous commuting block, digitized into  $r = 3$  rounds



**dispersed:**  $rm$  records reach the compiler. No record, and no proper prefix, determines the edge total  $x_j = \sum_t s_j^{(t)} a_j^{(t)} \bmod Q$ .

contrast  $c \cdot U^{2^0}$   $c \cdot U^{2^1}$   $c \cdot U^{2^2}$   $\times$  contiguous, but round 1 already fixes  $\theta_j(2^r - 1)$

## (b) Two currencies, at one cut



$$\overline{A}_p + (2p - 1) S \geq (1 - \delta) m \log_2 K_\epsilon - h_2(\delta)$$

plus a  $\log_2 \binom{m}{k}$  toll when the commitment schedule is private